Examiner only

10. Aqua regia is an acidic liquid that can dissolve unreactive metals such as gold. It can be formed by bubbling hydrogen chloride gas through concentrated nitric acid.

$$HNO_3(l) + 3HCl(g) \longrightarrow NOCl(g) + Cl_2(g) + 2H_2O(l) \qquad \Delta H^{\theta} = -69\ kJ\ mol^{-1}$$

Substance	Standard enthalpy change of formation, $\Delta_f H^{\theta}$ / kJ mol^{-1}	Standard entropy, S^{θ} / J K^{-1} mol^{-1}
$HNO_3(l)$	−173	156
$HCl(g)$		187
$NOCl(g)$	53	264
$Cl_2(g)$	0	223
$H_2O(l)$	−286	70

(a) Show that this is a redox reaction. [1]

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(b) State why the enthalpy change of formation of $Cl_2(g)$ is zero. [1]

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(c) Calculate the standard enthalpy change of formation of $HCl(g)$. [2]

$\Delta_f H^{\theta}$ = ... kJ mol^{-1}

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(d) (i) Calculate the free energy change, ΔG^{θ}, for this reaction at 25 °C. [3]

ΔG^{θ} = ... kJ mol^{-1}

(ii) An alternative method of preparing aqua regia is to use concentrated hydrochloric acid with concentrated nitric acid. State, giving a reason, the effect of this on the value of ΔG^{θ} at 25 °C. [1]

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(e) (i) The pH of a dilute solution of nitric acid is measured using a pH probe and has a value of 0.3. Calculate the concentration of this solution. [1]

Concentration = ... mol dm^{-3}

(ii) An alternative way of measuring the concentration of the solution is by titration against a standard solution of sodium hydrogencarbonate of concentration 0.500 mol dm^{-3}.

State and explain which of these two methods will give the more precise value for the concentration. [2]

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